

$$4 \quad (a) \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{20 \times 10^{-3}} = 314 \text{ rad s}^{-1}$$

$$B_0 = \frac{\mu I_0}{2\pi r} = \frac{(4\pi \times 10^{-7}) 4.0}{2\pi (0.15)} = 5.33 \times 10^{-6} \text{ T}$$

$$\begin{aligned} B &= B_0 \sin \omega t \\ &= \frac{\mu I_0}{2\pi r} \sin 314 t \\ &= \frac{(4\pi \times 10^{-7}) 4.0}{2\pi (0.15)} \sin 314 t \\ &= 5.33 \times 10^{-6} \sin 314 t \end{aligned}$$

(b) (i) The current in the cable creates a changing magnetic field

Since the solenoid experienced a change in magnetic flux linkage, an e.m.f. is induced in the solenoid.

$$\begin{aligned} (ii) \quad \varepsilon &= - \frac{d\Phi}{dt} \quad \text{where } \Phi = N B A \\ &= - N A \frac{dB}{dt} \quad \text{where } B = B_0 \sin \omega t \\ &= - N A (B_0 \omega \cos \omega t) \end{aligned}$$

$$\begin{aligned} V_{peak} &= (6000) (\pi 0.010^2) 5.33 \times 10^{-6} (314) \\ &= 3.15 \times 10^{-3} \text{ V} \end{aligned}$$

(iii) Half wave rectification over 2 cycles

Negative cosine graph and labels for voltage and period